Stage 1 - INAIO Sample Problems

Note:

1. Problems are either multiple choice questions with a single correct option or require numeric answers in the range 0 to 99.
2. In case of numeric answers, round the final answer downward to the nearest integer.
3. Use of Google, **free version** of ChatGPT, code execution, Python notebooks, calculators is permitted but the problems can be solved even without these tools and in fact, might often be faster and accurate.

Problem 1 [2pts]

A student is involved in three activities on a typical day: Robotics Club, Debate Team, and Sports Team. The chances of attending each activity on any single day are:

- Probability of attending Robotics Club is 50%
- Probability of attending Debate Team is 40%
- Probability of attending Sports Team is 60%

On any given day, the probabilities of attending pairs or all three activities together are:

- Probability of attending both Robotics and Debate: 30%
- Probability of attending both Robotics and Sports: 35%
- Probability of attending both Debate and Sports: 25%
- Probability of attending all three activities: 20%

Assume that the probability of attending each activity on a given day is the same for each day and is independent of attendance on other days.

Question: If the student did not attend Robotics on any of the four days from Monday to Thursday, what is the probability p that they still participated in the Sports Team on at least two of those days? Find $\lfloor 100p \rfloor$, the largest integer less than or equal to $100p$.

Problem 2 [2pts]

Four friends (Alice, Bob, Charlie, and David) are playing a coin exchange game where each player starts with 100 virtual coins. They exchange coins over several rounds, following a wealth distribution matrix below that determines how much each player keeps for themselves and how much they give to others.

Wealth Distribution Matrix:

	To Alice	To Bob	To Charlie	To David
From Alice	0.5	0.3	0.1	0.1
From Bob	0.2	0.6	0.2	0
From Charlie	0.1	0.2	0.5	0.2
From David	0	0.1	0.3	0.6

Each row represents how a player distributes their coins, including how much they keep for themselves (diagonal values). For example, the first row indicates that Alice keeps 50% of her coins, gives 30% to Bob, 10% to Charlie, and David each. **Note:** fractional coin exchanges are allowed.

Question: How many coins will the least wealthy person have after 2 rounds of exchanging coins? [Round downwards to closest integer]

Problem 3 [2pts]

Alex, Ben, and Cathy have discovered a treasure chest filled with coins. The treasure comes with three clues that will help them figure out how many coins each person should receive.
Clues:

1. If Alex gives 1 coin to Ben and 2 coins to Cathy, all three will have the same number of coins.
2. If Ben gives 1 coin to Alex and 5 coins to Cathy, Ben will have half as many coins as Cathy.
3. If Cathy gives 2 coins to Alex and 2 coins to Ben, Cathy will have half as many coins as Alex.

Question: How many coins does Ben receive?

Problem 4 [2pts]

You are a potion master trying to create a bottle of the weakest possible potion using two magical ingredients: Elf Dust and Dragon Powder. Both ingredients contribute to the potion's magic potency in a quadratic way.
Ingredient Contributions:

- Elf Dust contributes 2 points of strength per 1ml, 8 points per 2 ml and so on.
- Dragon Powder contributes 3 points of strength per ml, 12 points per 2 ml and so on.

Question: If the total volume of the potion has to be 100 ml, how much Elf Dust (in ml) should be there in the potion to minimize the potion's total magic strength? [Round downwards to closest integer]

Problem 5 [2pts]

Five friends—Alice, Ben, Chloe, David, and Emma—are each wearing a hat. The hats come in four colors: green, blue, yellow, and red. Each friend can see the hats of the others, but not their own. However, some friends have unique types of color blindness, which affects how they perceive certain colors.

Each friend's color blindness affects their statements. They can see the rest of colors clearly.

1. Alice: Can't distinguish green from blue (she sees both as "blue").
2. Ben: Can't distinguish red from yellow (he sees both as "red").
3. Chloe: Can't distinguish blue from yellow (she sees both as "yellow").
4. David: Can't distinguish green from red (he sees both as "red").

5. Emma: Can see all colors clearly.

The friends make the following observations based on what they can see:

1. Alice says: "I see one blue hat."
2. Ben says: "I see three red hats."
3. Chloe says: "I see two yellow hats."
4. David says: "Ben is wearing a blue hat and I see only one red hat."
5. Emma says: "Everyone is wearing a different color."

Question: Assuming all of them are truthful, what is the color of their hats in the order Alice, Ben, Chloe, David and Emma?

Options:

1. Blue, Green, Yellow, Red, Yellow
2. Blue, Green, Red, Yellow, Red
3. Green, Blue, Yellow, Red, Yellow
4. Green, Blue, Red, Yellow, Red
5. Red, Blue, Yellow, Green, Yellow
6. Yellow, Blue, Red, Green, Red

Problem 6 [2pts]

A player rolls a fair six-sided die repeatedly until they roll three 3's (not necessarily contiguously).
Question: What is the expected number of rolls required to get to the third 3?

Problem 7 [2pts]

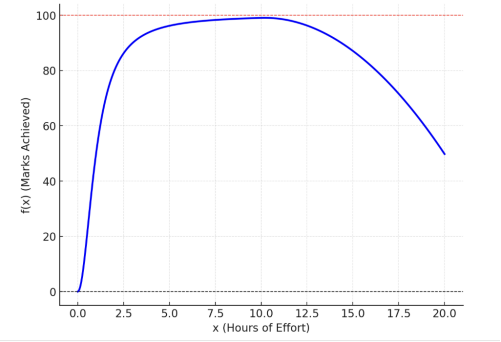


Figure 1: Variation of Marks $M(x)$ with Effort x

The plot above shows how the marks $M(x)$ achieved by a student vary with the number of hours of effort x. The function $\mathbf{1}_{[P]}$ is defined as:

$$\mathbf{1}_{[P]} = \begin{cases} 1 & \text{if the condition } P \text{ is true,} \\ 0 & \text{otherwise.} \end{cases}$$

Question: Which of the following equations best describes $M(x)$?
Options:

1. $M(x) = 100\left(1 - \frac{1}{1+x^2}\right)$
2. $M(x) = 100\left(1 - \frac{1}{1+x^2}\right) + (x-10)^2\,\mathbf{1}_{[x>10]}$
3. $M(x) = 100\left(1 - \frac{1}{1+x^2}\right) - 0.5(x-10)^2\,\mathbf{1}_{[x>10]}$
4. $M(x) = 100 - x^2$
5. $M(x) = 100\left(1 - \frac{x}{1+x^2}\right) - 0.3(x-8)^3\,\mathbf{1}_{[x>8]}$
6. $M(x) = 50\left(1 - \frac{1}{1+x^2}\right) - x + 0.2(x-12)^2\,\mathbf{1}_{[x>12]}$

Problem 8 [2pts]

Background: TF-IDF Ranking

TF-IDF (Term Frequency-Inverse Document Frequency) is a widely used method for retrieving relevant documents for a given query. It measures the relevance of terms by combining two key factors:

1. **Term Frequency (TF):** This measures how often a term (i.e., word) appears in a document, relative to the total number of words in the document. It is calculated as:
$$\text{TF(term)} = \frac{\text{Number of occurrences of the term in the document}}{\text{Total number of words in the document}}$$
2. **Inverse Document Frequency (IDF):** This assesses how unique or significant a term is across all documents in a collection. It is computed using the formula:
$$\text{IDF(term)} = \log_{10}\left(\frac{\text{Total number of documents}}{\text{Number of documents containing the term}}\right)$$
3. **TF-IDF Score:** The TF-IDF score of a term in a document is the product of its term frequency and inverse document frequency:
$$\text{TF-IDF(term)} = \text{TF(term)} \cdot \text{IDF(term)}$$

The overall TF-IDF score of a document is the sum of the TF-IDF scores of all query terms within that document. This score helps rank documents by their relevance to a given query.

You are tasked with determining the most relevant document for a given query, "artificial intelligence data" based on the table below. The relevance is measured using TF-IDF (Term Frequency-Inverse Document Frequency) scores.

Term	**Document 1**	**Document 2**	**Document 3**	**Document 4**
artificial	3	0	2	1
intelligence	1	2	0	1
data	2	1	3	5

Question: Calculate the TF-IDF scores for all documents and determine the score s of the highest-ranked document. Find $\lfloor 100s \rfloor$, the largest integer less than or equal to $100s$.

Problem 9 [15 pts]

Background: Binary Classification Metrics

The performance of binary classification (positive class vs. negative class) is commonly measured using the following metrics:

Confusion Matrix:

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Key Metrics:

- **Precision:** The fraction of correctly predicted positive instances among all predicted positive instances:
$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$
- **Recall:** The fraction of correctly predicted positive instances among all actual positive instances:
$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$
- **F1-Score:** The harmonic mean of precision and recall:
$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

The average utility of classifier is often computed by multiplying costs/gains associated with each cell of confusion matrix with the corresponding fraction.

A robot is designed to classify rubies from regular stones in various areas. While operating in a given area called Stone Valley where the Ruby-to-Stone Ratio is 1:4, it is observed that the robot has a Precision of 80% (0.8) and Recall of 75% (0.75). Further, for every input instances (stone), the mining robot makes gains (in INR) as per the following cost matrix:

	Predicted Ruby	Predicted Stone
Actual Ruby	10	-3
Actual Stone	-1	0

Questions:

Note: Round **final numeric answers** downward to the nearest integer.

1. Calculate the **F1-Score** for ruby detection in Stone Valley (in %). [1 pt]
2. Compute the Confusion Matrix for Stone Valley **(in %)**: [4 pts - 1 pt each]
   (a) Calculate the fraction of **True Positives (TP)**.
   (b) Calculate the fraction of **False Positives (FP)**.
   (c) Calculate the fraction of **True Negatives (TN)**.
   (d) Calculate the fraction of **False Negatives (FN)**.

3. Using the confusion and cost matrices, calculate the average utility for 10 instances in Stone Valley. [1 pt]

4. A new area, Ruby Ridge, has a Ruby-to-Stone ratio of **1:2**. Using the same robot classifier and cost matrix, calculate the average utility per 10 instances in this area. [4 pts]

5. For a third area, the classifier must achieve a total cost of **28 INR per 10 stones**. What should be the minimal fraction of rubies in this area to meet this requirement? [5 pts]
   Options:

   1. 10%
   2. 20%
   3. 30%
   4. 40%
   5. 50%
   6. 60%

Problem 10 [15 pts]

Background: Linear Regression and Regularization

Regression is a statistical method for modeling the relationship between a dependent variable (Y) and independent variables ($X_1, X_2, \ldots, X_n$) by estimating coefficients $\boldsymbol{\beta}$ that best explain the data.

Key Terms:

- $\mathbf{X}$: Represents all input features, including a column of ones for the intercept term. For m data points and n features:

$$\mathbf{X} = \begin{bmatrix} 1 & X_{1,1} & X_{2,1} & \cdots & X_{n,1} \\ 1 & X_{1,2} & X_{2,2} & \cdots & X_{n,2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{1,m} & X_{2,m} & \cdots & X_{n,m} \end{bmatrix}$$

- $\mathbf{Y}$: Actual outputs (dependent variable), expressed as:

$$\mathbf{Y} = \begin{bmatrix} Y_1 & Y_2 & \cdots & Y_m \end{bmatrix}^T$$

- $\boldsymbol{\beta}$: Coefficients, expressed as a vector:

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 & \beta_1 & \beta_2 & \cdots & \beta_n \end{bmatrix}^T$$

Regression Methods:

1. **Linear Regression:** Minimize the *Sum of Squared Errors (SSE)*:

$$J(\beta) = \sum_{i=1}^{m} \left(Y_i - (\beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_n X_{ni})\right)^2$$

   The closed-form solution is:

$$\boldsymbol{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}$$

2. **Ridge Regression:** An extension of previous method with an additional penalty term:

$$J(\beta) = \sum_{i=1}^{m} \left(Y_i - (\beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_n X_{ni})\right)^2 + \lambda \sum_{j=1}^{n} \beta_j^2$$

   The closed-form solution for Ridge Regression is:

$$\boldsymbol{\beta} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{Y}$$

   where λ is the regularization parameter, and $\mathbf{I}$ is the identity matrix with top right entry as zero.

You are tasked with predicting students' **Final Exam Scores** (Y) based on two features:

- X_1: Hours Studied
- X_2: Practice Tests Taken

The table below shows observations from multiple students.

Hours Studied (X_1)	Practice Tests (X_2)	Final Exam Score (Y)
1	2	50
2	4	60
3	5	65
4	6	70
5	8	75
6	9	80

Questions

Note: Round **final numeric answers** downward to the nearest integer.

1. Using **only** X_1 as the predictor in **Linear Regression**, compute: [2 pts]
   (a) Coefficient (β_1)
   (b) Intercept (β_0)
2. Using **both** X_1 **and** X_2 as predictors in **Linear Regression**, compute: [3 pts]
   (a) Coefficient (β_1)
   (b) Coefficient (β_2)
   (c) Intercept (β_0)
3. Compute the **SSE** using both X_1 and X_2 as predictors in the **Linear Regression** model. [2 pts]
4. Predict the likely score when $X_1 = 2.5$ and $X_2 = 3.5$ using the **Linear Regression** model with both predictors. [1 pts]
5. Assume X_2 values are changed to be exactly the same as X_1 for all rows. Use **Ridge Regression** with a regularization parameter $\lambda = 10$ and compute: [3 pts]
   (a) Coefficient (β_1)
   (b) Coefficient (β_2)
   (c) the intercept (β_0).
6. In the Ridge Regression problem of Q5, if the X_2 values are changed to be $-X_1$, which of the following is true about the new coefficients? [1 pt]
   Options:
   1. The coefficients (β_1) and (β_2) will both decrease in magnitude, moving closer to zero.
   2. The coefficient (β_2) will become zero.
   3. Only the intercept will change.
   4. All the coefficients will remain unchanged.
   5. The coefficient (β_1) will double in magnitude and coefficient (β_2) will be zero.
   6. The intercept will remain unchanged and the new coefficient (β_2) will be negative of the previous value.
7. In the Ridge Regression problem of Q5, if the X_2 values are changed to be all zero, which of the following is true about the new coefficients? [1 pt]
   Options:
   1. The coefficients (β_1) and (β_2) will both decrease in magnitude, moving closer to zero.
   2. The coefficient (β_2) will become zero but the coefficient (β_1) will be less than double the previous magnitude.

3. Only the intercept will change.
4. All the coefficients will remain unchanged.
5. The coefficient (β_1) will double in magnitude and coefficient (β_2) will be zero.
6. The intercept will remain unchanged.

8. In the Ridge Regression problem of Q5, if the X_2 values are changed to be $2X_1$, which of the following is true about the new coefficients? [1 pt]
Options:

1. The coefficients (β_1) and (β_2) will both decrease in magnitude, moving closer to zero.
2. New coefficients β_1 =2 β_2 and the intercept decreases
3. New coefficients β_2 =2 β_1 and intercept decreases.
4. Only the intercept will change.
5. All the coefficients will remain unchanged.
6. New coefficients β_2 =2 β_1 and the intercept remains unchanged.

9. In the Ridge Regression problem of Q5, if only the regularization parameter λ is increased to 20, which of the following is true about the new coefficients? [1 pt]
Options:

1. The coefficients (β_1) and (β_2) will decrease in magnitude, moving closer to zero while the intercept increases.
2. The coefficients (β_1) and (β_2) will increase in magnitude, moving further from zero.
3. Only the intercept will change.
4. All the coefficients will remain unchanged.
5. The coefficients (β_1) and (β_2) and intercept will all double in magnitude.
6. The coefficients (β_1) and (β_2) and intercept will all halve in magnitude.

Solutions

1. 68
2. 75
3. 16
4. 60
5. Option 3
6. 18
7. Option 3
8. 44
9. (1) 77
   (2) a) 15
   b) 3
   c) 76
   d) 5
   (3) 13
   (4) 23
   (5) Option 4
10. (1) (a) 5
   (b) 46
   (2) (a) 1
   (b) 2
   (c) 44
   (3) 8
   (4) 58
   (5) (a) 2
   (b) 2
   (c) 51
   (6) Option 6
   (7) Option 2
   (8) Option 3
   (9) Option 1